

[L6] Chapters 11 & 12 - Getting personal

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Part 2: More appropriation

- 1980s - 1990s, european culture was quite opposed to american influence and culture
- The squatter movement epitomizes that
- Many groups did not want to be associated with the US
- Europe and America had different mentalities regarding various topics
 - The Apple advertisement for the 1984 release of the Macintosh, referencing Orwell's "1984" novel, was more successful in europe than the US. Europe thought of the book as a novel about a society under constant surveillance, while the US viewed it as an anti-cold war book.
- The squatter movement was quite specific to europe (almost non-existent in the US)
 - Very active in europe (also politically active)
- While the Homebrew Computer Club stood against software piracy, the European Hobby Computer Club kept pirating software.
- Software prices were a problem, they wanted more affordable software
- The squatter movement gave birth to the concept of hacking as a political statement (hacktivism)
 - Hacking went from a childish thing to do to an actual crime, with Steffen Wernery being arrested for computer crimes
- Hacking also became a form of appropriation (something that the computer was not originally meant to do)

Demoscenes:

- Mostly european phenomenon 1980s
- Originated in the squatter culture, but not limited to it
- People started tinkering with the machine, trying to do things with it: ASCII art, graffiti, music
- This was distributed through diskmags that contained graphics, music, discussion of movies/music, etc
 - They borrowed their structure from journals

Free software movement:

- Rose from the academic scene as well as from the squatter culture
- Rose 1983 from MIT
 - This meant that, once bought, the user was free to use , thinker with and understand the software
- This was mostly picked up as meaning free = no cost
- The open source software movement originated from this (1998)

Personal Computers:

- The home computer projects (from 1986 onwards) made owning a computer at home more affordable, supported by the government
- In the netherlands: End of the 1980s: 30% owned a PC. 1995: 60% did
- They were mostly playing games on them
- Improvements in education were intended, getting the population more familiar with computer science
- With the advent of the internet more appropriation started taking place
- Events were organized (*Hacking at the end of the universe*, 1993, Netherlands)

- The digital city (De Digitale Stad) originated from this in 1994 (launched in Amsterdam). This was a place where you could digitally chat with people, go to the pub, post things to people, see books in the library, see forms that could be filled out in the town hall, etc
- This was very successful in Amsterdam. Within a week of the launch, all modems in the city were sold out. Lots of people wanted to be part of the digital city

Part 3: Rise of the internet

- As in the US, the internet in Europe was a confluence of free designs
- The wish for a worldwide encyclopedia was as much of a wish in Europe as in the US
- The desire for a fault tolerant network was also there (NATO)
- The desire to connect different networks was there too, but a politically funded wish, that was already there in the 1980s and late 1970s
- The french Minitel (1978 - 2012) promoted by France Telecom, was a way of promoting telephone connections at first
- It was sponsored heavily by the french government
- Like in the US, email was a source for success in Europe

In the Netherlands:

- 1980, Viditel was launched
- It was launched by the national telecommunications company
- There was not much content to be had
- It was quite expensive, hard to afford
- The project failed
- The other state-owned network, that of television and radio stations, launched a competing product: Teletext (1980)
- It came with the television set => anyone with a television could receive Teletext messages
- 1978, the dutch state-owned telecommunications company PTT had a network established, using circuit switching electronics: Datanet 1
- 1978-1981 they switched to packet switching on the X25-protocol
- It was launched through an existing network of telex-lines
- It cost hundreds of euros/month
- Because they shared lines, using datanet and the telephone simultaneously was not possible
- It was kept in use for a while because it was successful with those who could afford it: Banks, universities, large companies
- By the end of the 1980s, most people had several network lines, so occupying one with a computer was not that much of a sacrifice anymore
- The national networks could be both an advantage (France) and a disadvantage (Netherlands, where the competition between state companies proved a disadvantage)

Part 4: Rise of the game industry

- Playing as a game of appropriating
- 1952 Alan Turing designed a chess algorithm, but he didn't have a machine to run it on
- 1960, the game Nimbus was programmed by Regnecentralen to see if a program could match a human player
 - Used as a way to show what a computer could do
- 1961, Spacewar! was created by 2 MIT students on a DEC PDP-1

- 1972 The first game tournament took place
- In the early 70s, games were used for large arcade machines
- There was a growing culture of text-based adventure games (ex: Pong by Atari)
- A big change was Namco's 1980 Pacman release. It was released as software that you could buy on a cassette and play on your own computer. Completely different concept from selling a whole computer
- With lots of companies making games, sometimes very bad games, the game industry experienced a short dip.
- From that dip, it rose with better graphics, multiplayer options, greatly improved music
- Graphics and merchandise started being used extensively (ex: Resident Evil)
- Games started being used for educational purposes (ex: Kerbal Space Program)

Part 5: Rise of computer science / information science (as a profession)

- Professionalisation took place in the 1960s - 70s - 80s and continues to this day
- Academically, mathematics and engineering schools were the places where computer scientists would be educated in the 1960
- Management sciences were interested in computers from the perspective of automating offices
- In France, starting in the 1980s, computer science departments started being formed, usually as a mix between the engineering/mathematics departments and the management departments
- Roughly at the same time, Information Sciences are formed in the management schools and become schools of their own in the 1980s
- In the Netherlands, it was slightly different. The government had to be convinced to fund these departments
- The computer science departments appeared later (1981)
- Information Sciences only showed up in 1984 at the Tilburg Catholic School
- In Belgium and the UK, information sciences was part of management sciences. Lifestyle informatics/AI was part of the medicine department
- Other signs of professionalisations:
 - National and international organisations:
 - The mathematical centre
 - The SSAA
 - Already there, having been established in the late 1950s
 - Peer reviewed journals
 - Came quite late to the Netherlands (1970s) but already existed internationally, so people could publish.
 - In the Netherlands, they would usually publish in mathematics journals or in non-peer-reviewed journals
 - Jargon
 - Writing history
 - Already there in the 1970s
- The three traditions are connected roughly like this:
 - Administration => Information Sciences
 - Process Control => Artificial Intelligence
 - Science => computer Science
- After the war, these 3 groups really came together
- They benefitted from input from each other

- There is significant overlap
- With hindsight, establishing a connection and drawing lines is easy